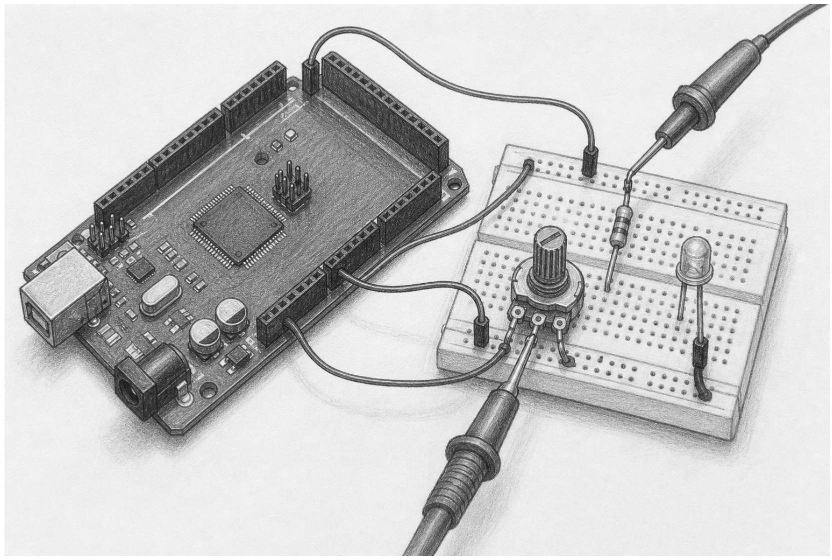
Turn a position into evidence

Lesson 007: `AnalogInput` and visible scaling

Predict → measure → sample → scale → display



Orientation plate. Use the graphite view to recognize the Mega, potentiometer, LED, and probe locations. It is not a wiring authority. The exact schematic and connection table below govern the circuit.

Question	Can one raw ADC sample remain observable while an explicit integer mapping drives PWM brightness?
Time	70–95 minutes
Prerequisites	Lessons 001, 004, and the resource-safe lifecycle
You need	Mega 2560, USB cable, breadboard, 10 kΩ linear potentiometer, ordinary LED, measured 220 Ω resistor, jumpers, and multimeter; logic analyzer or oscilloscope optional
Evidence	A0 wiper voltage, raw reading, D6 duty waveform, LED brightness, D13 acquisition indicator, and high-impedance shutdown
Safety class	E1: USB-powered Mega, passive potentiometer, and current-limited LEDs only
Status	Host-verified and experimental; physical Mega acceptance remains open

Safety and measurement boundary

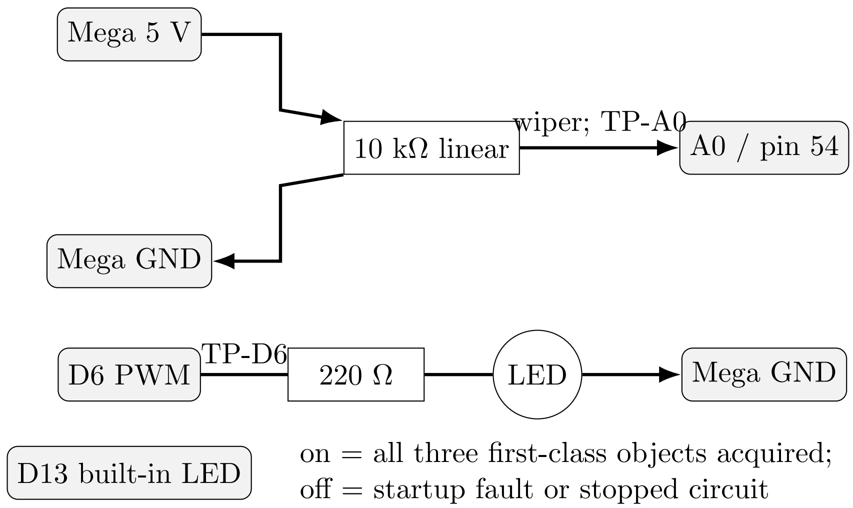
Use only the Mega's USB-derived 5 V and GND. The potentiometer divides that same measured supply, which is also the default ADC reference through AVCC; do not attach an external signal source or drive AREF. Remove USB before changing any wire. Put meter leads in the correct voltage sockets and clip the black lead to Mega GND while power is off. Use DC-voltage mode, never current mode.

Stop for heat, odor, smoke, USB disconnects, an unknown potentiometer terminal, voltage outside 0–5 V, or a meter lead in a current jack. Disconnect USB upstream and inspect the unpowered circuit.

Check — before wiring
☐ USB removed ☐ resistor measured ☐ potentiometer terminals identified ☐ meter in DC-voltage mode

1. Read the authoritative circuit

The potentiometer's outer terminals connect across 5 V and GND. Its center wiper is both the A0 input and named test point **TP-A0**. D6 drives a separate LED through its own 220 Ω resistor. D13 is the board's built-in diagnostic LED and consumes one claimed pin.



From	Through	To	Purpose
Mega 5 V	jumper	potentiometer outer 1	ADC reference-side terminal
potentiometer wiper	TP-A0	Mega A0	raw position input
potentiometer outer 2	jumper	Mega GND	common reference
Mega D6	TP-D6, 220 Ω	LED anode	PWM brightness output
LED cathode	jumper	Mega GND	output return
Mega D13	built-in circuit	built-in LED	acquisition indication

Unpowered proof. Verify that neither outer potentiometer terminal is shorted to its neighbor and that the LED path contains the resistor. Potentiometer rotation direction is not a safety property: swapping only the two outer terminals reverses the brightness direction.

2. Predict the evidence chain

The ADC reports an integer from 0 through 1023. With the default AVCC reference, one ideal code interval is $V_{\mathrm{ref}}/1024$, so

$$\text{reading} \approx \min\left(1023, \left\lfloor \frac{V_{\mathrm{TP-A0}}}{V_{\mathrm{ref}}} \times 1024 \right\rfloor\right).$$

The example maps that raw value to an 8-bit PWM duty:

$$\text{duty} = \left\lfloor \frac{\text{reading} \times 255 + 511}{1023} \right\rfloor.$$

The added 511 rounds rather than always truncating downward. All arithmetic is unsigned 32-bit so multiplication cannot overflow the AVR's 16-bit `int`.

Knob	Predicted TP-A0	Predicted raw	Predicted duty	Observed
low	0.0 V	0	0	
quarter	1.25 V	256	64	
middle	2.50 V	512	128	
three-quarter	3.75 V	767	191	
high	5.0 V	1023	255	

Code 1023 therefore represents the top interval below the reference, not an exact 5.000 V measurement. Real values vary with measured AVCC, potentiometer tolerance, contact noise, ADC quantization, converter error, and meter accuracy. The 10 kΩ potentiometer has at most about 2.5 kΩ Thevenin source resistance, comfortably below the datasheet's approximately 10 kΩ source-impedance guidance. Lesson 008 will calibrate and filter these measurements. This lesson deliberately preserves the raw sample.

3. Read the owning interface

`AnalogInput` is inert after construction. Initialization validates and claims the pin, selects input mode, and caches the first raw reading. `update()` performs one new conversion; `read()` returns that cached value without silently sampling again.

```
adk::Runtime     runtime;
adk::AnalogInput potentiometer (runtime.resources (), 54);

if (potentiometer.initialize ().ok ())
{
    potentiometer.update ();
    const adk::AnalogInput::Reading raw = potentiometer.read ();
}

potentiometer.shutdown ();
```

`Reading` is `uint16_t`; valid values are 0 through `AnalogInput::maximumReading` (1023). Pin 54 is Mega A0. Shutdown makes the pin high impedance, releases its claim, and retains the last cached reading only as historical software state. A retained number is not a live voltage.

`PwmOutput` owns D6. Its value is switched duty, not a steady analog voltage. Shutdown writes zero, selects input mode, and releases its pin and shared timer lease. `MonoLed` owns D13 and turns its semantic LED off before release.

4. Let the example tell the run-loop story

The canonical sketch declares the objects in dependency order. Its `setup()` acquires the diagnostic LED, input, and output. D13 turns on only after all three initialize. Its `loop()` then reads as the circuit does:

```
potentiometer.update ();
const adk::AnalogInput::Reading position   = potentiometer.read ();
const adk::PwmOutput::Duty      brightness =
    chooseBrightness (position);

if (!brightnessLed.write (brightness).ok ())
{
```

```
    stopSafely ();
}
```

Observe: take exactly one explicit raw sample. **Decide**: map position into the output domain. **Actuate**: command one PWM duty. Any command failure tears the whole circuit down; D13 turns off and both signal pins become high impedance.

The complete, compiled source is `examples/Lesson007AnalogInput/Lesson007AnalogInput.ino`. Do not maintain a second sketch copied from this page.

5. Build, upload, and observe from the CLI

No IDE is required. From the repository root:

```
make bootstrap
make arduino-Lesson007AnalogInput
make upload EXAMPLE=Lesson007AnalogInput PORT=/dev/ttyACM0
```

Use `make boards` to discover the port. This example intentionally emits no Serial records; its first evidence is the circuit. Repository targets still support later optional logs:

```
make monitor-describe PORT=/dev/ttyACM0
make monitor PORT=/dev/ttyACM0 BAUD=115200
make serial-log PORT=/dev/ttyACM0 BAUD=115200 \
    SERIAL_LOG=build/serial/lesson007.log
```

An empty Serial stream is expected here and is not a failure. Disconnect any monitor before uploading if the operating system cannot share the port.

6. Predict, observe, interpret

Resource-acquisition evidence

Predict: after reset, D13 remains off until every owner initializes, then stays on. **Observe**: watch D13 without touching the potentiometer. **Interpret**: on supports the claim that the program acquired A0, D6, and D13. It does not prove correct breadboard wiring or ADC accuracy.

Input evidence at TP-A0

Predict: with the meter's black lead on Mega GND, TP-A0 moves continuously from near 0 to near 5 V as the knob turns. **Observe**: measure at low, middle, and high positions. **Interpret**: the voltage proves the divider and common reference. It does not by itself prove which raw integer software cached.

Output evidence at TP-D6

Predict: the LED is off near raw 0 and grows brighter toward raw 1023. At middle position, a logic analyzer or oscilloscope shows D6 switching with about 50% high time. **Observe**: compare the LED and TP-D6. **Interpret**: the waveform supports correct scaling and PWM command. A handheld meter may display an average-like value; that reading is not proof of a steady analog voltage.

Safe-state evidence

After recording normal behavior, disconnect USB. Before rewiring, run a purpose-built shutdown acceptance sketch or host test that calls `shutdown()`; never short a live output to test it. Confirm D13 and the external LED turn off. With a suitable high-impedance instrument and the published acceptance procedure, verify A0 and D6 are no longer actively driven. This electrical observation is separate from the earlier D13 acquisition signal.

7. Diagnose by following the evidence

Observation	Most useful next check	What it separates
D13 stays off	remove power; inspect pin choices and unpowered continuity	initialization/resource failure from the run loop
D13 on, TP-A0 fixed	probe wiper, then both outer terminals relative to GND	divider wiring from software mapping
TP-A0 varies, LED fixed off	inspect LED polarity/resistor; then probe TP-D6	load wiring from PWM command
LED changes backward	swap the two outer potentiometer terminals while unpowered	orientation from correctness
LED flickers near one position	hold the knob; record TP-A0 stability	contact/sample noise from mapping defects
TP-D6 appears “2.5 V”	observe with a logic analyzer or oscilloscope	PWM average indication from a steady voltage

8. Deterministic host proof

The host fake supplies exact ADC readings; no test sleeps or depends on a physical knob. Required cases cover 0, 1, 511, 512, 1022, and 1023; invalid and non-analog pins; resource conflict; repeated initialization and shutdown; destruction while active; failed setup rollback; and pin reuse after release. The mapping table is pure integer arithmetic, so the same raw trace yields the same duties on every run.

```
make host-test
make host-test-exceptions
make quality
```

These commands prove software contracts and compilation, not the physical Mega circuit.

9. Bench acceptance card

Board / revision
Supply / measured 5 V
Arduino CLI / AVR core
Meter / probe
Sketch revision
Reviewer / date

Claim	Prediction	Observation	Interpretation
Resources acquired	D13 turns on after reset		
Low input	TP-A0 near 0 V; raw near 0; LED off		
Middle input	TP-A0 near half supply; duty near 128		
High input	TP-A0 near supply; duty 255		
PWM is switched	TP-D6 toggles at a middle setting		
Shutdown is safe	LEDs off; A0 and D6 high impedance		
Reset recovers	startup sequence repeats predictably		

Acceptance status: ☐ not run ☐ pass ☐ deviation recorded

10. Exercises

1. Calculate expected raw readings from your measured supply and TP-A0 values. Explain discrepancies without changing code.
2. Reverse only the two outer potentiometer wires while unpowered. Predict the entire evidence table before reconnecting USB.
3. Change the mapping so the minimum visible duty is 16 while raw zero still means off. State the new integer formula and boundary cases.
4. Add optional Serial records without changing the circuit-native acceptance criteria. Log raw and duty only when either changes by a useful threshold.
5. Sketch a calibration object for Lesson 008. Keep sampling, calibration, filtering, and PWM ownership as separate responsibilities.

Use with and next step

Use the HTML lesson for maintained API links, errata, source downloads, and copyable commands. Use this PDF at the bench for prediction, measurement, diagnosis, and the signed acceptance record. Lesson 008 adds calibration, dead bands, and deterministic filtering. Lesson 009 composes them into an adaptive night light with explicit sensor-fault states.

Primary references

- Arduino Mega 2560 Rev3 documentation and pinout
- ATmega2560 datasheet, ADC, and electrical characteristics

- Arduino CLI command reference

Deferred evidence. The repository records host and compile gates. No physical voltage, waveform, current, shutdown, reset, or power-removal claim becomes accepted until a person completes this card on the named Mega 2560 and publishes the result.